

GlobePay – SRS Version 1.0

1. Introduction

1.1 Purpose

The purpose of GlobePay is to provide international travelers with a simple and transparent payment experience by allowing them to:

- Scan existing merchant QR codes
- View real-time currency conversion
- Pay in their preferred currency
- Track international spending

GlobePay does not replace existing payment networks. It acts as an intelligent payment orchestration layer on top of existing payment infrastructures.

1.2 Product Vision

To become the preferred payment companion for international travelers by eliminating currency confusion and providing transparent cross-border payments.

1.3 Project Scope

The system will:

- ✓ Register travelers
- ✓ Authenticate users
- ✓ Link payment methods
- ✓ Scan merchant QR codes
- ✓ Detect payment information
- ✓ Fetch live exchange rates
- ✓ Convert currencies
- ✓ Display transaction costs
- ✓ Simulate payment processing
- ✓ Maintain transaction history

The system will not:

- ✗ Create a new payment network
 - ✗ Replace Visa/Mastercard
 - ✗ Provide merchant onboarding
 - ✗ Perform real bank settlement
 - ✗ Act as a licensed bank
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2. Stakeholders

Primary Stakeholders

Traveler

A user traveling internationally who wants to make payments in their preferred currency.

System Administrator

Manages the GlobePay platform.

External Stakeholders

Forex Rate Provider

Provides real-time exchange rates.

Payment Provider

Processes payments through existing payment infrastructures.

Banking Partner

Represents external banking systems.

3. User Roles

Traveler

Can:

- Register
- Login
- Complete profile
- Link payment methods

- Scan QR codes
 - View exchange rates
 - Make payments
 - View transaction history
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Administrator

Can:

- View users
 - Monitor transactions
 - View logs
 - Manage system settings
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4. Business Objectives

BO-001

Reduce confusion caused by foreign currencies.

BO-002

Provide transparent currency conversion.

BO-003

Enable international QR-based payments.

BO-004

Provide a unified traveler payment experience.

BO-005

Build a scalable fintech platform architecture.

5. Functional Requirements

Authentication Module

FR-001

User shall register an account.

FR-002

User shall login.

FR-003

User shall logout.

FR-004

User shall reset password.

Profile Module

FR-005

User shall update profile information.

FR-006

User shall upload profile picture.

FR-007

User shall manage linked payment methods.

QR Payment Module

FR-008

User shall scan QR codes.

FR-009

System shall extract payment information.

FR-010

System shall detect merchant currency.

Forex Module

FR-011

System shall fetch live exchange rates.

FR-012

System shall convert payment amounts.

FR-013

System shall display exchange rate details.

FR-014

System shall calculate estimated fees.

Payment Module**FR-015**

User shall confirm payment.

FR-016

System shall create payment transaction.

FR-017

System shall generate transaction ID.

FR-018

System shall store payment records.

Transaction Module**FR-019**

User shall view transaction history.

FR-020

User shall view transaction details.

FR-021

User shall download receipts.

Admin Module**FR-022**

Admin shall view users.

FR-023

Admin shall monitor transactions.

FR-024

Admin shall view system logs.

6. Non-Functional Requirements

Performance

NFR-001

System shall respond within 2 seconds.

NFR-002

System shall support 100,000 concurrent users.

Scalability

NFR-003

System shall support horizontal scaling.

NFR-004

System architecture shall support future growth to 10 million users.

Security

NFR-005

All communication shall use HTTPS.

NFR-006

Passwords shall be encrypted.

NFR-007

JWT authentication shall be used.

Availability

NFR-008

System availability shall be 99.9%.

7. Assumptions

- Exchange rate APIs are available.
- Merchant QR codes follow supported standards.
- Internet connectivity is available.

8. Constraints

- Currency rates fluctuate continuously.
 - Payment regulations vary by country.
 - Real bank integration is outside MVP scope.
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9. MVP Scope

Included

- User Authentication
- Profile Management
- QR Scanning
- Currency Conversion
- Payment Simulation
- Transaction History
- Admin Dashboard

Excluded

- Real Visa Integration
- Real Mastercard Integration
- Real SWIFT Transfers
- Merchant Portal
- Merchant Registration
- Regulatory Compliance Engine
- Fraud Detection Engine